

ClearEval: A Clinically Grounded, Physics-Informed Benchmark for Tissue Optical Clearing Protocol Design with Large Language Models

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Abstract. Tissue optical clearing (TOC) enables deep three-dimensional imaging of large biological specimens and is increasingly used in real laboratory workflows for neuroscience, developmental biology, and translational pathology. Designing a TOC protocol is a high-stakes, multi-objective decision process: small mistakes in incubation time, temperature, or reagent compatibility can irreversibly damage specimens. While large language models (LLMs) perform strongly on biomedical exams and static knowledge benchmarks, their ability to generate *physically executable* wet-lab protocols remains poorly measured, and common text-based metrics (e.g., BLEU/ROUGE) cannot detect fatal physical-chemical violations.

We introduce **ClearEval**, an expert-level benchmark for *end-to-end* TOC protocol design, built not only from curated literature but also from routine expert decision cases distilled into reusable templates. ClearEval mirrors the three critical stages of TOC practice—(i) objective specification, (ii) method selection, and (iii) protocol parameterization—and organizes the required knowledge into a three-layer taxonomy with nine subcategories and three difficulty levels. The benchmark comprises **1,208** multiple-choice questions for knowledge assessment and **70** open-ended protocol tasks with expert-written reference protocols. Beyond LLM-as-a-judge scoring for completeness and correctness, we propose **physics-informed effectiveness metrics** that verify refractive-index matching and diffusion-limited time feasibility, together with hard constraints for fluorescence compatibility. Benchmarking diverse proprietary and open-source LLMs reveals a pronounced *knowledge-action gap*: models often succeed at static knowledge retrieval but fail to generate protocols that satisfy hard physical constraints. Our effectiveness score correlates strongly with blinded expert ratings (Pearson $r > 0.85$), whereas semantic metrics show weak correlation ($r < 0.4$), supporting ClearEval as a practical testbed for clinically grounded scientific assistants.

Keywords: Benchmark · Tissue optical clearing · Large language models · Protocol generation · Physics-informed evaluation

1 Introduction

Large language models (LLMs) have demonstrated impressive performance in biomedical question answering and clinical exam-style benchmarks. However, real-world clinical translation requires more than fluent text: in wet-lab workflows, a single incorrect parameter (e.g., incubation time, temperature, solvent choice) can cause protocol failure or irreversibly damage valuable specimens. This motivates benchmarking LLMs on tasks that reflect *how experts actually work*—from goals, to method choice, to parameterized protocols—and evaluating outputs against *hard physical and chemical constraints* rather than surface-level textual similarity.

Tissue optical clearing (TOC) provides a timely and representative testbed. TOC reduces scattering and absorption, enabling deep volumetric imaging and supporting the construction of three-dimensional anatomical and pathological atlases. In practice, TOC protocol design is a multi-objective optimization problem balancing transparency (refractive index matching), fluorescence preservation (chemical compatibility), morphology preservation, safety, and time/cost constraints. This cross-disciplinary reasoning requirement makes TOC suitable for probing LLMs’ scientific “theory-to-practice” capability.

Why existing benchmarks and metrics fall short. Current scientific benchmarks largely measure static recall and short-form reasoning, while protocol-generation evaluation often relies on BLEU/ROUGE or pure LLM judging. Such metrics can miss *fatal but fluent* errors—e.g., incompatible solvents or diffusion-impossible incubation times—that matter for safe deployment in real workflows.

Our solution. We propose **ClearEval**, the first benchmark tailored to end-to-end TOC protocol design, emphasizing physical executability and expert workflow realism. ClearEval is constructed via an expert-automation pipeline grounded in (i) daily expert decision cases distilled into templates and structured knowledge entries, and (ii) curated literature evidence used for retrieval-augmented data generation.

Contributions.

1. **Benchmark:** ClearEval models the three critical decision stages of TOC practice and provides a three-layer taxonomy (tissues & markers, reagents, methods) with nine subcategories and three difficulty levels.
2. **Dataset:** We release 1,208 multiple-choice questions (MCQ) and 70 open-ended protocol tasks (OEQ) with expert reference protocols, created via a rigorous expert-automation pipeline with global semantic de-duplication.
3. **Evaluation:** We introduce physics-informed and hard-constraint metrics for protocol *effectiveness*, complementing LLM-as-a-judge scoring for completeness and correctness.
4. **Findings:** We benchmark diverse LLMs and reveal a consistent knowledge-action gap; physics-informed effectiveness aligns with human experts, while semantic metrics do not.

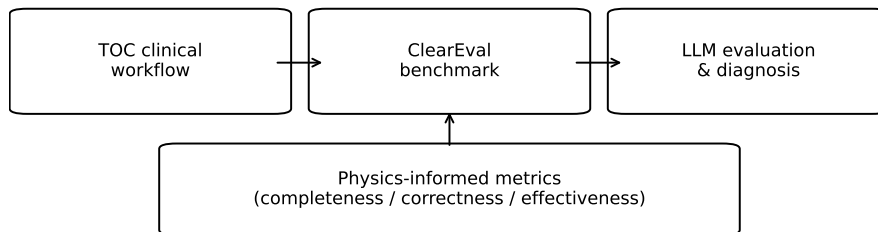


Fig. 1. ClearEval is designed for theory-to-practice evaluation: it mirrors the TOC clinical workflow and uses physics-informed metrics to assess executability.

2 Related Work

2.1 Scientific Benchmarks Beyond Static Knowledge

Broad scientific benchmarks (e.g., multi-task multiple-choice exams) have driven progress in scientific language understanding, but their formats primarily emphasize recall and short-form reasoning. Recent benchmarks begin to evaluate scientific workflows and planning, yet measuring *wet-lab executability* remains challenging.

2.2 Evaluation for Protocol Generation

Standard NLG metrics (BLEU/ROUGE) quantify lexical overlap and are insufficient for scientific protocols. LLM-as-a-judge has gained popularity but may suffer from “hallucination resonance” when both generator and judge overlook physical infeasibility. ClearEval combines LLM judging (for structure and logical validity) with physics-informed verification (for feasibility).

3 The ClearEval Benchmark

3.1 TOC Workflow: Three Expert Decision Stages

ClearEval mirrors the expert TOC workflow: (i) specify the imaging objective and targets (tissue and marker choices), (ii) select a clearing method under constraints (fluorescence compatibility, tissue properties, time/cost), and (iii) customize a step-by-step protocol with concrete parameters (temperature, concentration, incubation time).

3.2 Task Taxonomy and Difficulty Levels

We organize TOC knowledge into three pillars—**tissues & markers**, **reagents**, and **clearing methods**—and nine subcategories. To probe capability boundaries, tasks are stratified into three difficulty levels: **L1** named entity recognition, **L2** domain understanding, and **L3** practical problem solving (full protocol design).

3.3 Dataset Construction from Literature and Daily Expert Cases

ClearEval is built through a three-stage pipeline: schema design, data generation, and post-processing. In schema design, experts distilled **60** template questions from routine TOC decision cases and created a structured knowledge base. We additionally curated **30** core TOC papers and segmented them into **437** labeled evidence chunks for retrieval. In data generation, a retrieval-augmented workflow produced an initial pool of **2,000** MCQs and ~ 80 OEQs. In post-processing, we applied global semantic de-duplication (cosine similarity > 0.85) followed by model-assisted verification and full expert review, yielding **1,208** MCQs and **70** OEQs.

4 Evaluation

ClearEval evaluates LLMs along two complementary axes: **knowledge** (MCQ) and **application** (OEQ).

4.1 MCQ: Knowledge Evaluation

We report accuracy and macro-F1 across categories to reflect class imbalance.

4.2 OEQ: Protocol Quality Assessment

Protocol quality is assessed along: **completeness** (structural coverage and parameter granularity), **correctness** (logical order, reagent–method consistency, plausibility), and **effectiveness** (quantitative feasibility verification).

Physics-informed effectiveness. We evaluate:

$$S_{\text{trans}} = \exp \left(-\frac{(\text{RI}_{\text{reagent}} - \text{RI}_{\text{tissue}})^2}{2\sigma_{\text{RI}}^2} \right), \quad (1)$$

$$S_{\text{time}} = \exp \left(-\frac{(\Delta t)^2}{2\sigma_t^2} \right), \quad (2)$$

where S_{trans} penalizes refractive-index mismatch and S_{time} penalizes deviation from an empirically feasible time interval motivated by diffusion constraints. We further apply a hard constraint S_{label} for fluorescence compatibility: any incompatibility yields zero score.

Table 1. Main leaderboard on ClearEval. Replace placeholder numbers with your finalized results.

Model	$I_K \uparrow$	$I_A \uparrow$	ExpertScore $\uparrow$	Setting
Model A	–	–	–	1-shot
Model B	–	–	–	RAG

4.3 Overall Score

We compute a balanced **Expert Score** using the harmonic mean of the normalized knowledge index I_K and application index I_A :

$$\text{ExpertScore} = \frac{2I_K I_A}{I_K + I_A}. \quad (3)$$

5 Experiments

5.1 Models and Settings

We benchmark diverse proprietary and open-source LLMs (e.g., GPT-, Gemini-, Claude-, GLM-, Qwen-, DeepSeek-, and MiniMax-family models) under: **MCQ**: zero-shot and one-shot prompting; **OEQ**: one-shot and retrieval-augmented generation (RAG). For RAG, we retrieve $\text{top-}k = 5$ evidence chunks from the ClearEval knowledge base. We set temperature to 0.7 for all models.

5.2 Main Results: A Knowledge–Action Gap

Table ?? summarizes the main leaderboard. Even top models achieve high knowledge scores but exhibit a sharp drop on application, indicating a consistent knowledge–action gap in protocol design.

5.3 Human Alignment of Effectiveness Metrics

We evaluate whether physics-informed effectiveness aligns with expert judgments. Across 100 sampled protocols, effectiveness scores correlate strongly with blinded expert ratings (Pearson $r > 0.85$), while ROUGE-L correlates weakly ($r < 0.4$), indicating that semantic overlap fails to capture protocol executability.

5.4 Fine-Grained Diagnosis and the Limits of Retrieval

A category-wise analysis reveals spiky capability profiles: models perform well on high-level method selection but struggle with full-process design and troubleshooting. RAG improves completeness but yields limited gains on physics-grounded effectiveness (e.g., time feasibility), suggesting that retrieval is necessary but not sufficient for physical grounding.

6 Discussion: Clinical Translation and Global Health Relevance

ClearEval is designed to support MICCAI’s “theory-to-practice” trajectory: it evaluates whether LLMs can generate protocols that satisfy hard constraints required for safe execution. By grounding tasks in routine expert decision cases and anchoring evaluation to physics-informed checks, ClearEval provides actionable signals for improving scientific assistants beyond text fluency.

Toward real-world workflows. A practical deployment of protocol assistants requires reliable detection of fatal errors, traceable evidence, and integration with laboratory SOPs. ClearEval can serve as a pre-deployment gatekeeper: models that score well must pass both expert-judged structure and physics-grounded feasibility checks.

Global accessibility. Protocol expertise is unevenly distributed across institutions. Benchmarks that measure executable protocol design can help develop assistants that reduce the barrier to adopting advanced 3D imaging and sample preparation pipelines in diverse settings. Future extensions will incorporate broader case distributions (e.g., different organisms, tissues, and resource constraints) and prospective evaluations in real lab environments.

7 Conclusion

We presented ClearEval, an expert-level, clinically grounded benchmark for TOC protocol design with physics-informed evaluation. ClearEval reveals a pronounced knowledge–action gap in current LLMs and provides metrics that align with expert judgments. We hope ClearEval will facilitate the development of reliable scientific agents for 3D biomedical imaging workflows.